

Waleed Alghaithi

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Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, Linux
- **Software Engineering:** OOP, data structures & algorithms, software design, debugging, unit testing, Jest, Playwright
- **AI & Data Foundations:** Linear algebra, calculus, probability, NumPy, Pandas, LangGraph, LiteLLM
- **Developer Tools:** Git, GitHub, BitBucket, Jira, Linux/Unix, VS Code, Cursor, JetBrains IDEs, command-line tools, Docker, Grafana
- **Web & Databases:** HTML, CSS, React, Node.js, FastAPI, SQL/MySQL, SQLite, RESTful APIs

Experience

Undergraduate Researcher, McNair Scholars Program

University of California San Diego

San Diego, CA

Feb 2026 – Present

- Selected for the McNair Scholars Program, a federally funded initiative preparing first-gen and underrepresented students for doctoral study through faculty-mentored research.
- Conducting research on spatial reasoning failures in vision-language models, analyzing internal expert activations in Qwen3-VL-30B across the LEGO-Puzzles benchmark.
- Running evaluation pipelines on RunPod GPU infrastructure, comparing model performance across height, position, rotation, and ordering spatial tasks to identify where and why models fail.

Tutor and Teaching Assistant

City College of San Francisco

San Francisco, CA

Jan 2023 – May 2025

- Tutored 20+ students weekly in Physics and CS, covering topics from Kinematics and Circuits to Data Structures, Algorithms, and OOP in Java and Python.
- Developed reusable study materials and worked closely with instructors to align tutoring sessions with course expectations.
- Boosted average exam scores by 15% and maintained a 90% retention rate among regular attendees.

Machine Learning Research Intern

Lawrence Berkeley National Laboratory

Berkeley, CA

June 2023 – August 2023

- Processed raw mmWave radar binaries into structured CSV datasets by decoding headers against JSON configs and validating outputs against Infineon radar visualizations.
- Built and evaluated SVM and K-means models for snowpack and precipitation classification, achieving 90%+ accuracy across multiple weather conditions.
- Optimized and deployed the full pipeline on Raspberry Pi (Linux), achieving sub-second inference for low-power remote environmental sensing.

Projects

Spatial Reasoning Failures in Vision-Language Models | Python, Qwen3-VL, RunPod, NumPy, Pandas Jan 2026 – Present

- Analyzed why Qwen3-VL-30B struggles at spatial reasoning by tracking which internal experts activate on correct vs. incorrect answers.
- Ranked experts by how strongly their activation correlated with correct answers across height, position, rotation, and ordering tasks.
- Benchmarked Qwen3-VL against GPT-4o and Gemini Flash 2.0 on RunPod to compare failure patterns across models.

Watchtower:

Observability Platform | JavaScript, Node.js, SQLite, HTML, CSS, Clerk, Jest, Playwright, GitHub Actions Apr 2026 – Jun 2026

- Built a lightweight Sentry-like observability platform that captures JS errors, performance metrics, and user interactions through a browser SDK and displays them on a real-time dashboard.
- Designed the shared event schema and API contract used across all prototypes, standardizing how events are captured, stored in SQLite, and served via a Node.js backend.

- Researched and implemented Clerk for authentication and evaluated Resend and Gmail API for email alerting on high-risk events.
- Multi-Agent Travel Coordinator** | *Python, Multi-agent Systems, Docker, Cisco AGNTCY, FrontEnd, Git*

Feb 2026
- Built a 4-agent travel planning system using FastAPI and LangGraph, hitting 90%+ accuracy on natural language requests with zero scheduling conflicts.
 - Deployed with Docker and Cisco AGNTCY for agent communication, pulling real-time flight and hotel data via LiteLLM and SerpAPI.
 - Designed a React 19 and Tailwind CSS frontend for planning and viewing multi-leg itineraries in one dashboard.
- Precipitation Classification on Embedded Radar** | *Python, Linux, NumPy, Pandas, BitBucket*

Aug 2023
- Built an mmWave radar pipeline that automated raw binary processing, cutting preprocessing time by 60%.
 - Trained SVM and K-means models for precipitation classification, achieving 90%+ accuracy.
 - Deployed on Raspberry Pi with sub-second inference for low-power remote sensing.

Education

Bachelor of Science in Computer Science	University of California San Diego
<i>Relevant Coursework: Algorithms, Software Engineering, Advanced Data Structures, ML: Learning Algorithms, AI: Search & Reasoning, Deep Reinforcement Learning, Systems Programming, Linear Algebra</i>	<i>Anticipated Jun 2027</i>
<i>GPA: 3.7/4.0</i>	
Associate of Science in Computer Science	City College of San Francisco
<i>GPA: 4.0/4.0</i>	<i>Aug 2022 - May 2025</i>